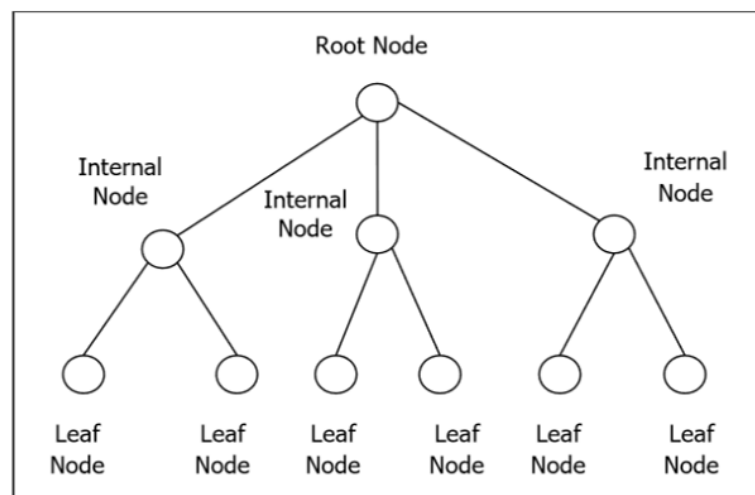


Trees

In discrete structure, a tree is a connected, acyclic graph used to represent hierarchical relationships. It consists of nodes (or vertices) connected by edges, with a unique path between any two nodes.

- **Nodes/Vertices:** The points or circles in the structure. The node with no parent is called the root.
- **Edges:** The lines connecting the nodes.
- **Acyclic:** The graph contains no cycles or circuits, meaning there is only one way to get from any one node to another.
- **Connected:** There is a path between any two nodes in the graph.
- **Leaves:** Nodes with a degree of 1 (only one edge connecting to them).
- **Internal Nodes:** Nodes with a degree of 2 or more.
- **Relationship to graphs:** A tree is a specific type of graph. A graph with no cycles that is not connected is called a forest, which is a collection of trees.



Common types of trees

- **Rooted Tree:** A tree with a designated root node.
- **Binary Tree:** A rooted tree where each node has at most two children.
- **Binary Search Tree:** A binary tree that maintains an ordered structure, allowing for efficient searching.
- **Spanning Tree:** A tree that includes all vertices of a graph.
- **Decision Tree:** A tree used in decision analysis to map out potential outcomes.

To solve for the number of edges (E) in a tree, use the property that an undirected tree with vertices (V) has exactly:

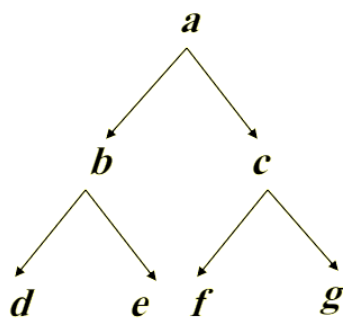
$$E = V - 1$$

Problem: A tree has 100 vertices. How many edges does it have?

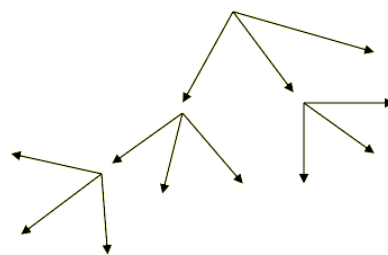
Solution: In a tree, the number of edges (E) is one less than the number of vertices (V) So, $E = V - 1 = 100 - 1 = 99$

m – ary trees

A rooted tree is called m-ary if every internal vertex has no more than m children. It is called full m-ary if every internal vertex has exactly m children. A 2-ary tree is called a **binary tree**.



Full binary tree



Full 3-ary tree

These problems rely on a set of core formulas: $n = mi + 1$ (*total vertices*), $n = mL - 1 / m - 1$ (*total vertices given leaves*), and $i = L - 1 / m - 1$ (*internal nodes given leaves*).

For example, a full 3-ary tree with 100 vertices would have 25 leaves, calculated by $L = (m - 1) i + 1$, which is which is $L = (3 - 1) (100 - 1) / 2 + 1 = 25$.

Formulas:

- Given internal nodes (i):
 - Total vertices (n): $n = mi + 1$
 - Leaves (L): $L = (m - 1)i + 1$
- Given total vertices (n):
 - Internal nodes (i): $i = \frac{n - 1}{m}$
 - Leaves (L): $L = \frac{(m - 1)n + 1}{m}$
- Given leaves (L):
 - Total vertices (n): $n = \frac{mL - 1}{m - 1}$
 - Internal nodes (i): $i = \frac{L - 1}{m - 1}$ ⓘ